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Analysis of Case Studies, MAPREDUCE, Fault Tolerance

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Amazon Cloud

- AWS stands for Amazon Web Services.
- The AWS service is provided by the Amazon that uses distributed IT infrastructure to provide different IT resources available on demand.
- It provides different services such as infrastructure as a service (IaaS), platform as a service (PaaS) and packaged software as a service (SaaS).
- Amazon launched AWS, a cloud computing platform to allow the different organizations to take advantage of reliable IT infrastructure.

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Advantages of AWS

1) Flexibility

- We can get more time for core business tasks due to the instant availability of new features and services in AWS.
- It provides effortless hosting of legacy applications.
- AWS also offers a choice that whether we want to run the applications and services together or not.
- We can also choose to run a part of the IT infrastructure in AWS and the remaining part in data centres.

2) Cost-effectiveness

- AWS requires no upfront investment, long-term commitment, and minimum expense when compared to traditional IT infrastructure that requires a huge investment.

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3) Scalability/Elasticity

- Through AWS, auto scaling and elastic load balancing techniques are automatically scaled up or down, when demand increases or decreases respectively.
- AWS techniques are ideal for handling unpredictable or very high loads. Due to this reason, organizations enjoy the benefits of reduced cost and increased user satisfaction.

4) Security

- AWS provides end-to-end security and privacy to customers.
- AWS has a virtual infrastructure that offers optimum availability while managing full privacy and isolation of their operations.

Amazon Elastic Compute Cloud (EC2)

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- It is a virtual server platform allowing users to create and run virtual machines on Amazon Server firm.
- The Amazon Machine Images (AMI) is used by EC2 to launch and run server instances for running operating systems like: Linux (Red-Hat), Windows, etc. on different servers.

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- A number of tools are used to support Amazon's web services.

Amazon Simple Queue Service is a message queue and transaction system for distributed Internet-based applications.

Amazon Simple Notification Service is used to publish message from an application.

Amazon CloudWatch is used for monitoring EC2 Cloud which supports by providing console or command line view of resources in utilization.

Elastic Load Balancing is used to detect whether an instance is failing or check whether the traffic is healthy or not.

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- Based on the concept of Pay-As-You-Go, AWS provides the services to the customers.
- You can scale the compute capacity up and down as per the computing requirement changes.
- Implementing a service might require the following components:

• Application server (having large RAM allocation)
• A load balancer
• Database server
• Firewall and network switches
• Additional rack capacity

EC2 Pricing Options

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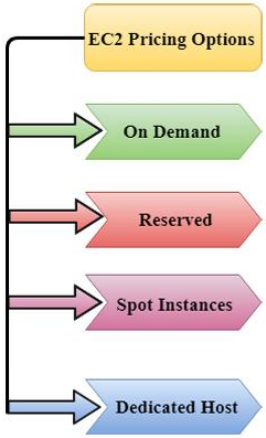

- On Demand

- It allows you to pay a fixed rate by the hour or even by the second with no commitment.
- Linux instance is by the second and windows instance is by the hour.
- On Demand is perfect for the users who want low cost and flexibility of Amazon EC2 without any up-front investment or long-term commitment.

- Reserved

- It is a way of making a reservation with Amazon or we can say that we make a contract with Amazon.
- The contract can be for 1 or 3 years in length.
- In a Reserved instance, you are paying some upfront, so it gives you a significant discount on the hourly charge for an instance.
- It is used for those applications that require reserved capacity.

- Spot Instances

- It allows you to bid for a price whatever price that you want for instance capacity, and providing better savings if your applications have flexible start and end times.
- Spot Instances are useful for those applications that have flexible start and end times.
- It is useful for those applications that are feasible at very low compute prices.

- Dedicated Hosts

- A dedicated host is a physical server with EC2 instance capacity which is fully dedicated to your use.
- The physical EC2 server is the dedicated host that can help you to reduce costs by allowing you to use your existing server-bound software licenses.

Simple Storage Service (S3)

- Bucket Data, in S3, is stored in containers called buckets. Each bucket will have its own set of policies and configurations.
- This enables users to have more control over their data.
- Bucket Names must be unique.
- There is a limit of 100 buckets per AWS account. But it can be increased if requested by AWS support.
- The maximum size of an AWS S3 bucket is 5TB. It consists of the following:

1. Key.
2. Version ID.
3. Value.
4. Metadata.
5. Sub resources.
6. Access control information.
7. Tags.

- S3 Versioning

- Versioning means always keeping a record of previously uploaded files in S3.
- Versioning keeps all the copies of your file, so, it adds cost for storing multiple copies of your data.
- Versioning is helpful to prevent unintended overwrites and deletions.

- Bucket policy

- A document for verifying the access to S3 buckets from within your AWS account, controls which services and users have what kind of access to your S3 bucket.
- Each bucket has its own Bucket Policies.

- Access control lists (ACLs)

- A document for verifying access to S3 buckets from outside your AWS account.
- An ACL is specific to each bucket.

- Lifecycle Rules:

- This is a cost-saving practice that can move your files to AWS Glacier (The AWS Data Archive Service) or to some other S3 storage class for cheaper storage of old data or completely delete the data after the specified time.

- Key:

- The key, in S3, is a unique identifier for an object in a bucket.

Advantages of S3 Bucket

- Scalability:

- Amazon S3 can be scalable horizontally which makes it handle a large amount of data. It can be scaled automatically without human intervention.

- High availability:

- S3 bucket is famous for its high availability nature you can access the data whenever you required it from any region. It offers a Service Level Agreement (SLA) guaranteeing 99.9% uptime.

- Data Lifecycle Management:

- You can manage the data which is stored in the S3 bucket by automating the transition and expiration of objects based on predefined rules. You can automatically move the data to the Glacier, after a specified period.

- Integration with Other AWS Services:

- You can integrate the S3 bucket service with different services in the AWS.

Amazon RDS

- Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate and scale relational databases in the cloud.
- Amazon Relational Database Service (RDS) is a managed SQL database service provided by Amazon Web Services (AWS).
- Amazon RDS supports an array of database engines to store and organize data.
- It also helps in relational database management tasks like data migration, backup, recovery and patching.

Amazon RDS Features

- Replication:

- RDS uses the replication feature to create read replicas, and these are read-only copies of the database instances that the application uses without changing the original production database.

- Monitoring:

- The Amazon CloudWatch service enables managed monitoring, and it lets users view capacity and I/O metrics.

- Patching:

- RDS provides patches for whichever database engine the user chooses.

- Backups:

- RDS provides managed instance backups with transaction logs to enable point-in-time recovery.

Benefits of Amazon RDS

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- Easy to administer:
  - Amazon RDS makes it easy to go from project conception to deployment.
- Highly scalable:
  - You can scale your database's compute and storage resources with only a few mouse clicks often with no downtime.
- Available and durable:
  - Amazon RDS runs on the same highly reliable infrastructure used by other Amazon Web Services.
- Fast:
  - Amazon RDS supports the most demanding database applications.
- Secure:
  - Amazon RDS makes it easy to control network access to your database.

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- Inexpensive:
  - You pay very low rates and only for the resources you consume.
- Ease of use:
  - Admins don't need to learn specific database management tools, and they also can manage multiple database instances using the management console.
- Cost-effectiveness:
  - According to AWS, customers only pay for what they use.

Drawbacks of Amazon RDS

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- Lack of root access:
  - Because it is a managed service, users do not have root access to the server running RDS.
- Downtime:
  - Systems must go offline for some patching and scaling procedures, and the timing of these processes varies.

AWS Cloud Development Kit (AWS CDK)

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- The AWS CDK lets you build reliable, scalable, cost-effective applications in the cloud with the considerable expressive power of a programming language.

- Benefits includes:

- **Easier cloud onboarding**
  - Developers can use their language of choice and continue using their preferred integrated development environment (IDE) to write a CDK app.
- **Faster development process**
  - The expressive power of programming languages and features, can significantly accelerate the development process.

- **Customizable and shareable**
- CDK allows you to extend existing components to create custom components that meet your organization's security, compliance, and governance requirements.
- These components can be easily shared around your organization.

- **No context switching**
- You can write your runtime code and define your AWS resources with the same programming language, and can continue using the same IDE for runtime code and infrastructure development.

Google Cloud Platform(GCP)

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- GCP is a cloud computing platform that allows users to build, deploy, and scale applications, websites, and services using Google-managed data centers.
- It provides infrastructure, storage, data analytics, artificial intelligence, networking, and developer tools that run on the same infrastructure Google uses internally for its own products (like YouTube, Gmail, and Google Search).
- It provides:

IaaS (Infrastructure as a Service) – virtual machines, storage, and networking.

PaaS (Platform as a Service) – app development and deployment platforms.

SaaS (Software as a Service) – ready-to-use software applications.

Key Services of Google Cloud Platform (GCP)

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- Compute Services

- Used to run applications and workloads on Google's infrastructure.

- Compute Engine: Virtual Machines (VMs) for flexible computing.

- App Engine: Platform for developing and hosting web apps automatically.

- Cloud Run: Deploy and run containerized apps easily without managing servers.

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- Storage Services

- Used to store and manage data efficiently.

- Cloud Storage: Object storage for unstructured data (e.g., images, videos).

- Persistent Disk: Block storage for VMs.

- Filestore: Managed file storage for applications.

- Cloud Storage Buckets: For data archiving and backup.

- Database Services

- Used for structured and unstructured data management.

- Cloud SQL: Managed relational databases (MySQL, PostgreSQL, SQL Server).

- Cloud Firestore / Datastore: NoSQL document databases for mobile and web apps.

- Bigtable: High-performance NoSQL database for large analytical workloads.

- Spanner: Global, horizontally scalable relational database.

- Networking Services

- Used to connect and deliver applications globally.

- Virtual Private Cloud (VPC): Private, secure network for your resources.

- Cloud Load Balancing: Distributes user traffic efficiently.

- Cloud CDN (Content Delivery Network): Delivers content quickly worldwide.

- Cloud DNS: Scalable and reliable domain name system.

- Big Data & Analytics Services

- Used for data processing, analysis, and visualization.

- BigQuery: Serverless data warehouse for fast SQL analytics.

- Dataflow: Stream and batch data processing.

- Pub/Sub: Messaging service for real-time event streaming.

- Artificial Intelligence (AI) & Machine Learning

- Used to build and deploy AI models.

- Vertex AI: Unified platform for ML model training and deployment.

- TensorFlow: Open-source ML framework.

- AI APIs: Pre-trained models for Vision, Speech, Natural Language, and Translation.

- Developer & Management Tools

- Used for building, monitoring, and managing applications.

- Cloud SDK: Command-line tool for managing GCP resources.

- Cloud Build: for building and deploying code.

- Cloud Source Repositories: Private Git repositories.

- Cloud Monitoring & Logging: Observability tools for app health and performance.

- Identity & Security Services

- Used to protect applications and data.

- Cloud IAM (Identity and Access Management): Manage user access and roles.

- Cloud Security Command Center: Centralized security management.

- Cloud KMS (Key Management Service): Manages encryption keys.

- Shielded VMs & VPC Service Controls: Enhance system and data protection.

Advantages of GCP

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- Scalability – Automatically scales resources up/down based on usage.
- Global Infrastructure – Data centers across the world.
- Security – Built-in encryption, IAM, and compliance standards.
- Cost Efficiency – Pay-as-you-go pricing and sustained use discounts.
- AI & ML Integration – Access to Google's advanced AI models.
- Sustainability – Runs on 100% renewable energy.

GCP Security Features

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- IAM (Identity and Access Management) – Control who can access what.
- VPC Service Controls – Prevent data exfiltration.
- Encryption – Data encrypted at rest and in transit.
- Cloud Armor – DDoS protection.

Google Compute Engine

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- Google Compute Engine is an Infrastructure as a Service (IaaS) that lets users run virtual machines (VMs) on Google's cloud infrastructure.

- These VMs can run Linux or Windows operating systems, and users can configure:

- Number of CPUs
- Amount of RAM
- Disk size and type (SSD/HDD)
- Network and firewall settings

Key Features of Compute Engine

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Feature	Description
Custom Machine Types	You can choose the exact number of vCPUs and memory instead of fixed configurations.
Live Migration	VMs can be moved to another host without rebooting (useful for maintenance).
Snapshots & Images	Back up VM disks or create custom machine images for reuse.
Autoscaling	Automatically adds or removes VM instances based on traffic load.
Load Balancing	Distributes incoming network traffic across multiple VMs.
GPU/TPU Support	Attach GPUs or TPUs for high-performance computing or AI/ML tasks.

Components of Compute Engine

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- Instances – Virtual machines you create and run.
- Machine Types – Defines CPU, memory, and performance class.
- Images – OS templates (e.g., Ubuntu, Windows Server, Debian).
- Disks – Persistent or local storage.
- Networks – Virtual Private Cloud (VPC) networks that connect VMs.
- Firewalls – Rules that control inbound and outbound traffic.
- Metadata – Key-value pairs that provide configuration data to instances.

Advantages

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- Highly customizable machine types
- Scalable and flexible
- Global network performance
- Cost-effective pricing (with sustained use discounts)
- Integration with other GCP services (like BigQuery, Cloud Storage, etc.)

Google Developer Tools

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- Google Developer Tools are a collection of software, SDKs, APIs, and cloud services that help developers:

- Build, test, and deploy web or mobile applications
- Manage code and collaboration
- Monitor performance
- Integrate Google services like Maps, Firebase, Ads, AI, etc.
- They cover everything from frontend web development to backend cloud infrastructure.

Popular Google Tools

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Tool	Used For
Firebase	Fast app backend, database, hosting, analytics.
Android Studio	Building Android apps.
Flutter	Cross-platform app UI toolkit.
Chrome DevTools	Debugging and optimizing web apps.
TensorFlow / Vertex AI	Machine learning and AI.
Cloud SDK	Manage GCP via command line.
BigQuery	Analyze huge datasets in seconds.

Benefits of Google Developer Tools

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- Scalable and reliable infrastructure
- Integration with Google services
- Security and privacy controls
- Global developer community and documentation

Database

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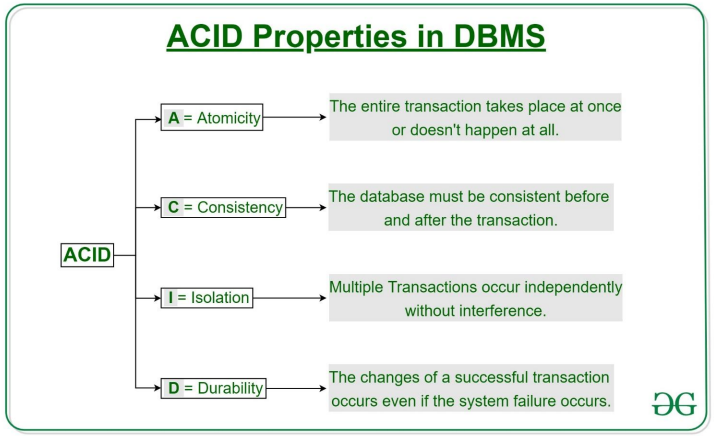
- In relational databases information is stored in tables, rows and columns, which typically works best for structured data.
- They offer ACID consistency mode for the data, which means:



Cloud Storage

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- Google Cloud Storage (GCS) is a service provided by Google Cloud Platform (GCP) that allows users to store and access data (files, backups, media, etc.) on Google's infrastructure. It's a highly scalable, durable, and secure object storage system — suitable for both individuals and enterprises.
- Google Cloud Storage is an object storage service for storing unstructured data such as images, videos, documents, logs, and backups.
- Data is stored in "buckets", which are containers that hold your objects (files).

Key Features

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Feature	Description
Object Storage	Stores data as objects (not as files or blocks). Each object contains data, metadata, and a unique ID.
Scalability	Scales automatically to handle any amount of data.
Durability	99.999999999% (11 nines) annual durability due to multi-region replication.
Availability	High availability with automatic failover.
Security	Data encrypted at rest and in transit; supports IAM and access control lists (ACLs).
Versioning	Keeps multiple versions of an object for backup and recovery.
Lifecycle Management	Automatically moves data between storage classes or deletes old data based on rules.
Public or Private Access	You can make data public (for websites) or keep it private (for backups).

Cloud tools for Eclipse

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- Google Cloud Tools for Eclipse is a plugin developed by Google that allows Java developers to build, deploy, and manage applications on Google Cloud Platform (GCP) directly from the Eclipse IDE.
- It integrates Eclipse with GCP services like App Engine, Cloud Storage, Cloud Pub/Sub, and more, helping developers easily move from local development to cloud deployment.

Supported Google Cloud Services

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Service	Purpose / Integration
App Engine	Build and deploy scalable web apps.
Compute Engine	Manage and deploy Java apps on virtual machines.
Cloud Storage	Connect applications to Google Cloud Storage for file storage.
Cloud SQL	Integrate with relational databases in GCP.
Pub/Sub	Add message-based communication between microservices.

Benefits of Using Cloud Tools for Eclipse

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- Simplifies cloud deployment — no need to leave Eclipse
- Provides local testing and debugging before deployment
- Offers code templates and wizards for cloud projects
- Integrates with Google Cloud SDK for command-line compatibility
- Easy to monitor logs and performance inside the IDE
- Reduces configuration effort for Java developers moving to the cloud

Map Reduce

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- MapReduce is a framework using which we can write applications to process huge amounts of data, in parallel, on large clusters in a reliable manner.
- The MapReduce algorithm contains two important tasks, namely Map and Reduce.
- Map takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key/value pairs).
- Secondly, reduce task, which takes the output from a map as an input and combines those data tuples into a smaller set of tuples.
- As the sequence of the name MapReduce implies, the reduce task is always performed after the map job.

Architecture

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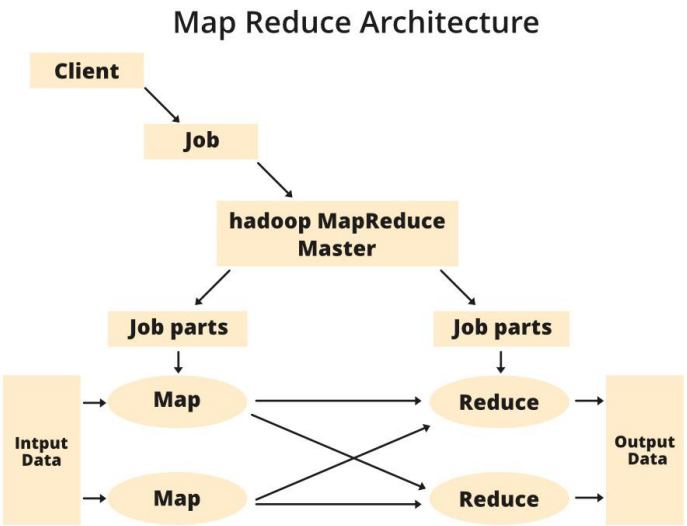

Components of MapReduce Architecture

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- **Client:** The MapReduce client is the one who brings the Job to the MapReduce for processing. There can be multiple clients available that continuously send jobs for processing to the Hadoop MapReduce Manager.
- **Job:** The MapReduce Job is the actual work that the client wanted to do which is comprised of so many smaller tasks that the client wants to process or execute.
- **Hadoop MapReduce Master:** It divides the particular job into subsequent job-parts.
- **Job-Parts:** The task or sub-jobs that are obtained after dividing the main job. The result of all the job-parts combined to produce the final output.
- **Input Data:** The data set that is fed to the MapReduce for processing.
- **Output Data:** The final result is obtained after the processing.

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- In MapReduce, we have a client.

- The client will submit the job of a particular size to the Hadoop MapReduce Master.

- Now, the MapReduce master will divide this job into further equivalent job-parts.

- These job-parts are then made available for the Map and Reduce Task.

- This Map and Reduce task will contain the program as per the requirement of the use-case that the particular company is solving.

- The input data which we are using is then fed to the Map Task and the Map will generate intermediate key-value pair as its output.

- The output of Map i.e. these key-value pairs are then fed to the Reducer and the final output is stored on the HDFS.

How MapReduce Works

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- Suppose you want to count the number of times each word appears in a set of documents.
- **Step 1: Input Splitting**

- The input file (say, a large text) is split into smaller **chunks** (blocks).

- Each block is processed **in parallel** by different nodes in the cluster.

- **Step 2: Mapping Phase**

- Each node runs the **Map function** on its chunk.

- The function processes data and produces **key-value pairs**. Example:

- Input: "cat dog cat"
- Output: ("cat", 1)
  ("dog", 1)
  ("cat", 1)

- **Step 3: Shuffling and Sorting**

- The system automatically groups together all values with the **same key** across all Mappers.
Example:

("cat", [1,1])

("dog", [1])

- **Step 4: Reducing Phase**

- Each Reducer processes all values for a given key and aggregates them.
Example:

Reduce("cat", [1,1]) → ("cat", 2)

Reduce("dog", [1]) → ("dog", 1)

- **Step 5: Output**

- The results are written to an output file:

cat 2

dog 1

Application Of MapReduce

- **Entertainment:**

- To discover the most popular movies, based on what you like and what you watched in this case Hadoop MapReduce help you out. It mainly focuses on their logs and clicks.

- **E-commerce:**

- Numerous E-commerce suppliers, like Amazon, Walmart, and eBay, utilize the MapReduce programming model to distinguish most loved items dependent on clients' inclinations or purchasing behavior.

- It examines website records, buy history, user interaction logs, etc.

- **Data Warehouse:**

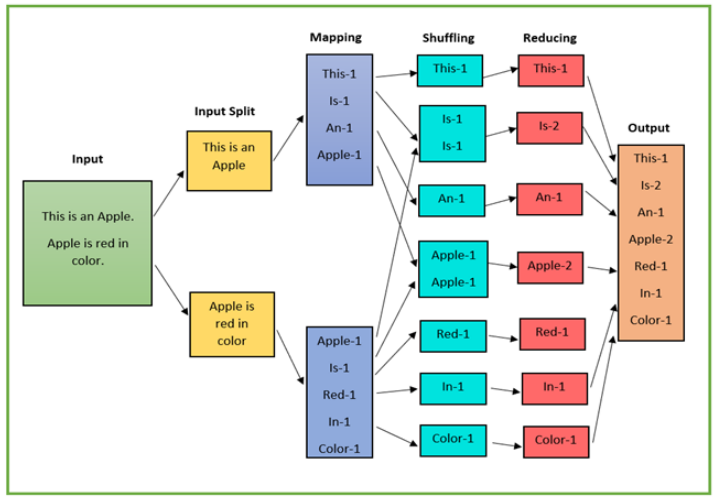
- We can utilize MapReduce to analyze large data volumes in data warehouses while implementing specific business logic for data insights.

- **Fraud Detection:**

- MapReduce is utilized in monetary enterprises, including organizations like banks, insurance providers, installment areas for misrepresentation recognition, or business metrics through transaction analysis.

Map Reduce Example

- The text is, "This is an apple. Apple is red in color."



Hadoop

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- **Hadoop** is an **open-source framework** developed by the **Apache Software Foundation** for **storing and processing large-scale data** in a **distributed computing environment**.
- It allows you to **store massive amounts of data** across many computers and **process it in parallel** efficiently and reliably.

Main Components of Hadoop Ecosystem

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Component	Description
1. HDFS (Hadoop Distributed File System)	A distributed file system that stores large files by splitting them into blocks and replicating them across multiple nodes.
2. YARN (Yet Another Resource Negotiator)	Manages and schedules system resources for running applications in the Hadoop cluster.
3. MapReduce	The data processing model — divides tasks into Map and Reduce phases.
4. Hadoop Common	A collection of utilities and libraries required by other Hadoop modules.

3 Types of Schedulers in Hadoop

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- **FIFO Scheduler**

- First In First Out, so the tasks or application that comes first will be served first.
- This is the default Scheduler we use in Hadoop.
- The tasks are placed in a queue and the tasks are performed in their submission order.
- In this method, once the job is scheduled, no intervention is allowed.
- So sometimes the high-priority process has to wait for a long time since the priority of the task does not matter in this method.

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- **Advantage:**

- No need for configuration
- First Come First Serve
- simple to execute

- **Disadvantage:**

- Priority of task doesn't matter, so high priority jobs need to wait
- Not suitable for shared cluster

- **Capacity Scheduler**

- Introduced by Yahoo!.
- Divides the cluster into **multiple queues**, each with a **configured capacity (percentage)** of cluster resources.
- Each queue can run **multiple jobs simultaneously**.

- **Advantage:**

- Best for working with Multiple clients or priority jobs in a Hadoop cluster.
- Maximizes throughput in the Hadoop cluster.

- **Disadvantage:**

- More complex.
- Not easy to configure for everyone.

- **Fair Scheduler**

- The Fair Scheduler is very much similar to that of the capacity scheduler.
- The priority of the job is kept in consideration.
- Developed by Facebook.
- Ensures that all jobs **get an equal share** of cluster resources **over time**.
- If one job finishes, the resources are redistributed to others.

- **Advantages:**

- Resources assigned to each application depend upon its priority.

- **Disadvantages:**

- The configuration is required.

Fault Tolerance in Hadoop

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- Fault tolerance refers to the ability of a system to recover from failures and errors without losing data or functionality.
- Hadoop relies on data replication and checkpointing to ensure fault tolerance, which means that it duplicates the data blocks across multiple nodes and periodically saves the state of the computation to disk.
- This provides high reliability and durability, but also consumes more disk space and network bandwidth.
- Hadoop is best for low-cost high-availability environments that can tolerate disk I/O and network latency.

Example of HDFS Fault Tolerance

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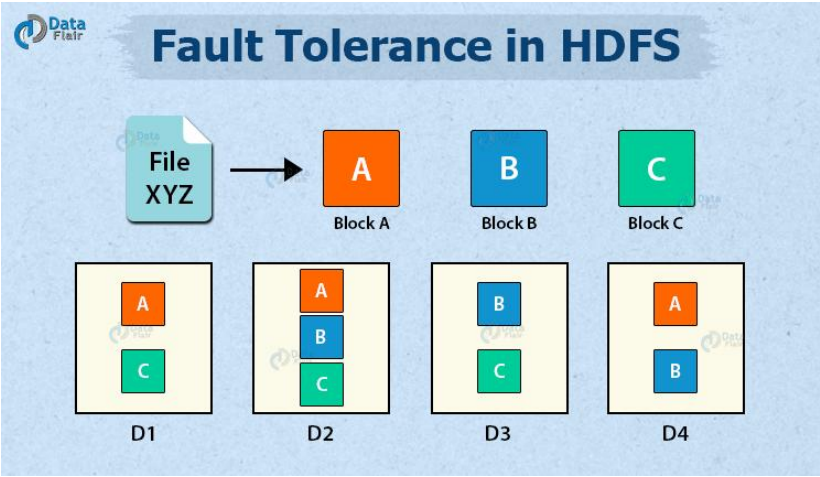

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- Suppose the user stores a file XYZ.
- HDFS breaks this file into blocks, say A, B, and C.
- Let's assume there are four DataNodes, say D1, D2, D3, and D4.
- HDFS creates replicas of each block and stores them on different nodes to achieve fault tolerance.
- For each original block, there will be two replicas stored on different nodes.
- Let the block A be stored on DataNodes D1, D2, and D4, block B stored on DataNodes D2, D3, and D4, and block C stored on DataNodes D1, D2, and D3.
- If DataNode D1 fails, the blocks A and C present in D1 are still available to the user from DataNodes (D2, D4 for A) and (D2, D3 for C).
- Hence even in unfavorable conditions, there is no data loss.